Regularization Techniques

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Regularization Techniques in Machine Learning

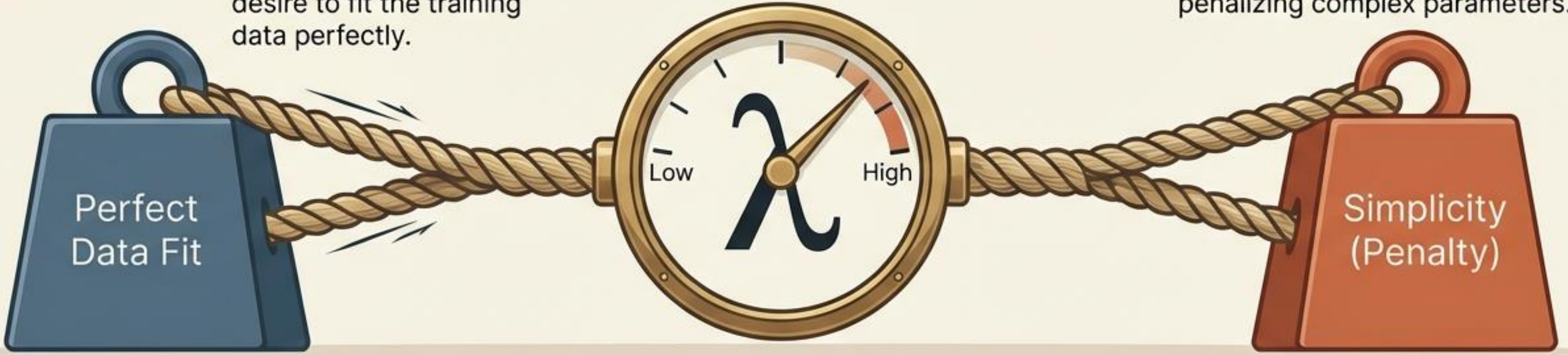
Understanding How to Prevent Overfitting with a Simple Example

The universal law of regularization is a tug-of-war

$$\text{New Loss} = \text{Data Loss} + \lambda \times \text{Penalty}$$

Data Loss: The model's desire to fit the training data perfectly.

Penalty: A structural constraint penalizing complex parameters.



λ: The control dial for regularization strength.

L2 Regularization shrinks weights toward zero

Best for: Stable weights, variance reduction, and handling many correlated features.

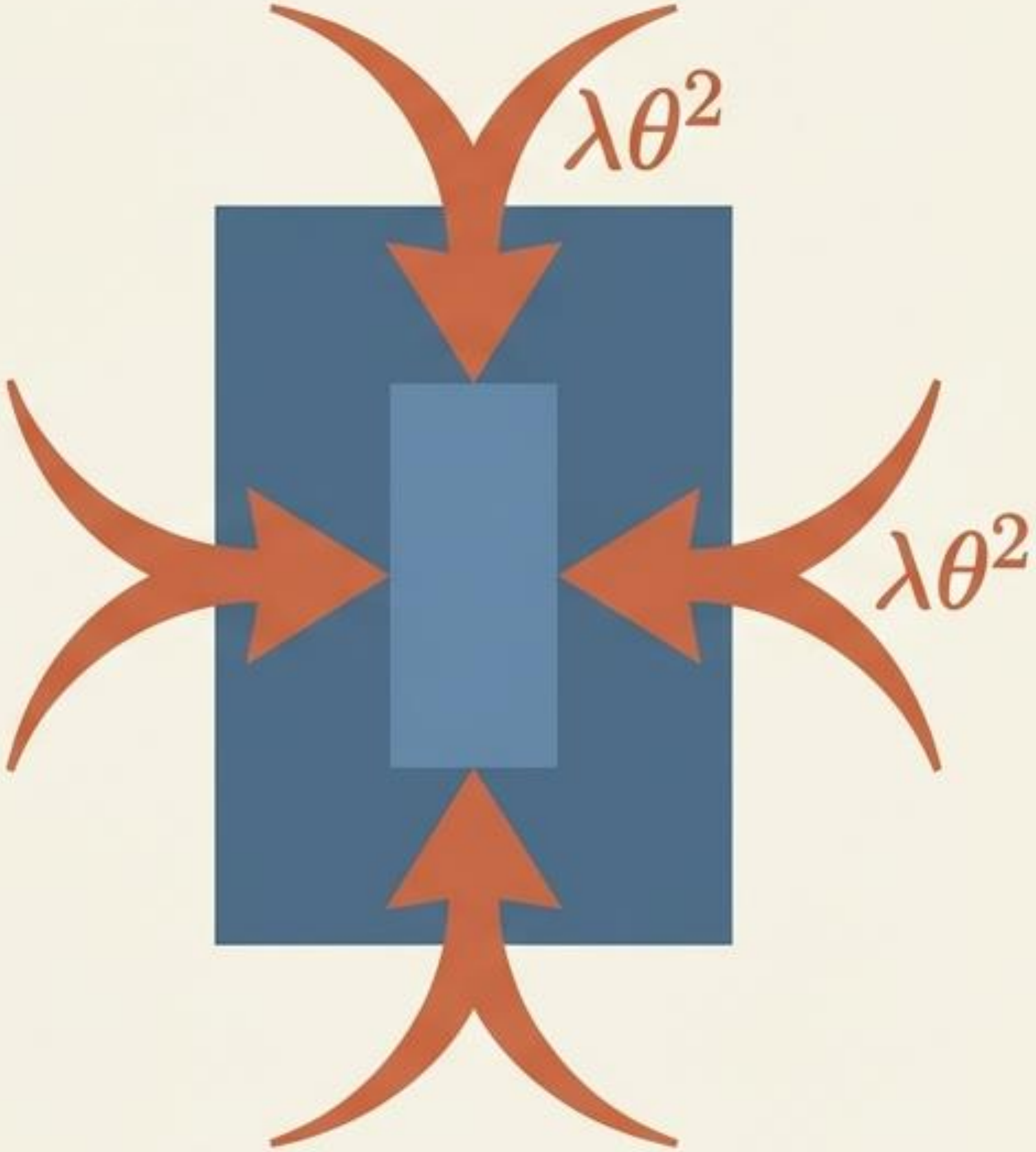
Loss Function:

$$J(\theta) = \frac{1}{m}\sum \ell(\hat{y}_i, y_i) + \lambda \sum \theta_j^2$$

Update Rule:

$$\theta \leftarrow \theta - \alpha(\nabla_\theta \, \text{DataLoss} + 2\lambda\theta)$$

Takeaway: L2 smoothly shrinks weights toward 0, resulting in a smoother model, but rarely makes them exactly 0.



Key Regularization Techniques Overview

L1 Regularization (Lasso)

Adds the sum of absolute weights as a penalty term. Promotes sparsity by driving some weights exactly to zero, effectively performing feature selection.

L2 Regularization (Ridge)

Adds the sum of squared weights as a penalty. Shrinks all weights smoothly toward zero without eliminating any features entirely—maintaining model stability.

Dropout

Randomly deactivates neurons during training to prevent co-adaptation. Forces the network to learn robust features independently rather than relying on specific neuron combinations.

Early Stopping

Monitors validation error during training and stops when performance starts deteriorating. Prevents the model from over-optimizing to training noise.

How L1 and L2 Regularization Work

Algorithm 0: The Core Idea (One Sentence)

Train the model to fit the data AND penalize large/complex parameter

$$\textbf{New Loss} = \textbf{Data Loss} + \boldsymbol{\lambda} \times \textbf{Penalty}$$

- λcontrols regularization strength.

Algorithm 1: L2 Regularization (Ridge / Weight Decay)

When to use

- You want **stable weights**, reduce variance
- Many correlated features

Loss

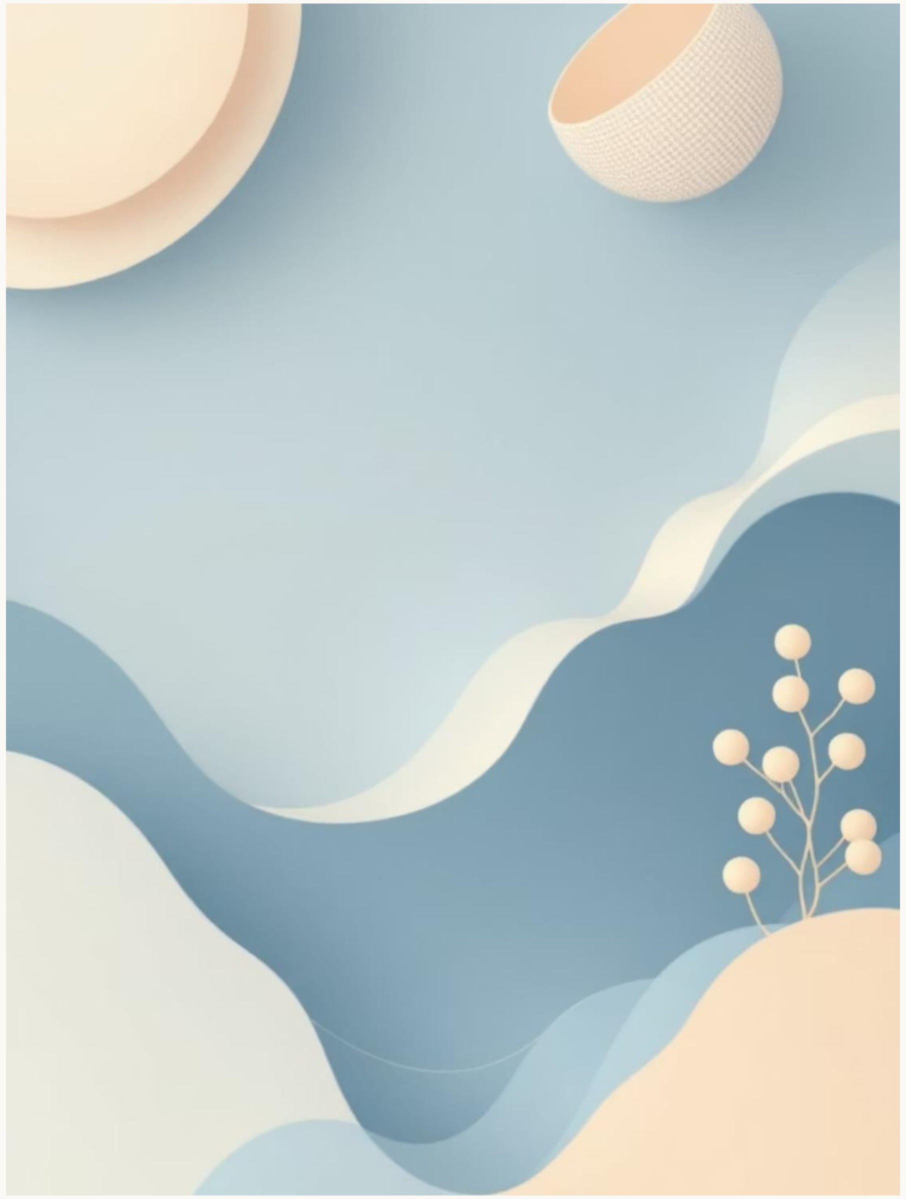
$$\boldsymbol{J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell(\hat{y}_i, y_i) + \lambda \sum_{j=1}^{n} \theta_j^2}$$

(usually exclude bias $\theta_0$from penalty)
Update rule (Gradient Descent form)

$$\theta \leftarrow \theta - \alpha(\nabla_\theta \boldsymbol{DataLoss} + 2\lambda\theta)$$

Interpretation

- L2 **shrinks** weights toward 0 (but rarely makes them exactly 0)



How L1 and L2 Regularization Work

Algorithm 2: L1 Regularization (Lasso)

When to use

- You want **feature selection** (sparse model)
- You suspect many features are irrelevant

Loss

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m} \ell(\hat{y}_i, y_i) + \lambda \sum_{j=1}^{n} |\theta_j|$$

Key concept

- L1 can make many weights **exactly zero** → selects features

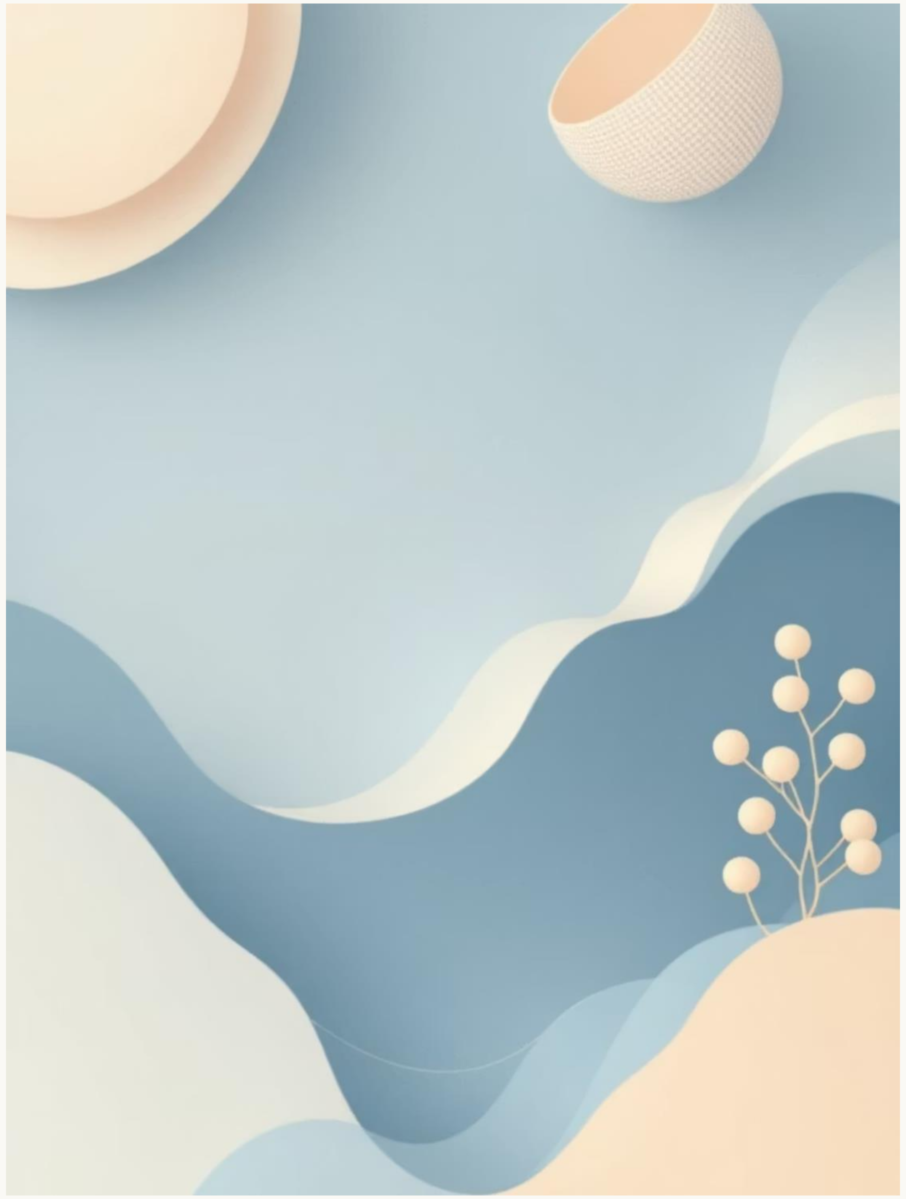
Practical update (simple teaching version)

Use **subgradient**:

$$\theta \leftarrow \theta - \alpha(\nabla_\theta \boldsymbol{DataLoss} + \lambda\, \boldsymbol{sign}(\theta))$$

where sign(0)can be treated as 0.

(Advanced: proximal/soft-thresholding, but you can keep it at this level.)



How L1 and L2 Regularization Work

Algorithm 3: Elastic Net (L1 + L2)

When to use

- You want **both** sparsity (L1) and stability (L2)

Loss

$$J(\theta) = \boldsymbol{DataLoss} + \lambda_1 \sum |\theta_j| + \lambda_2 \sum \theta_j^2$$

Algorithm 4: Dropout (Neural Networks)

When to use

- Deep learning models
- Many parameters → high overfitting risk

Training rule

For each mini-batch:

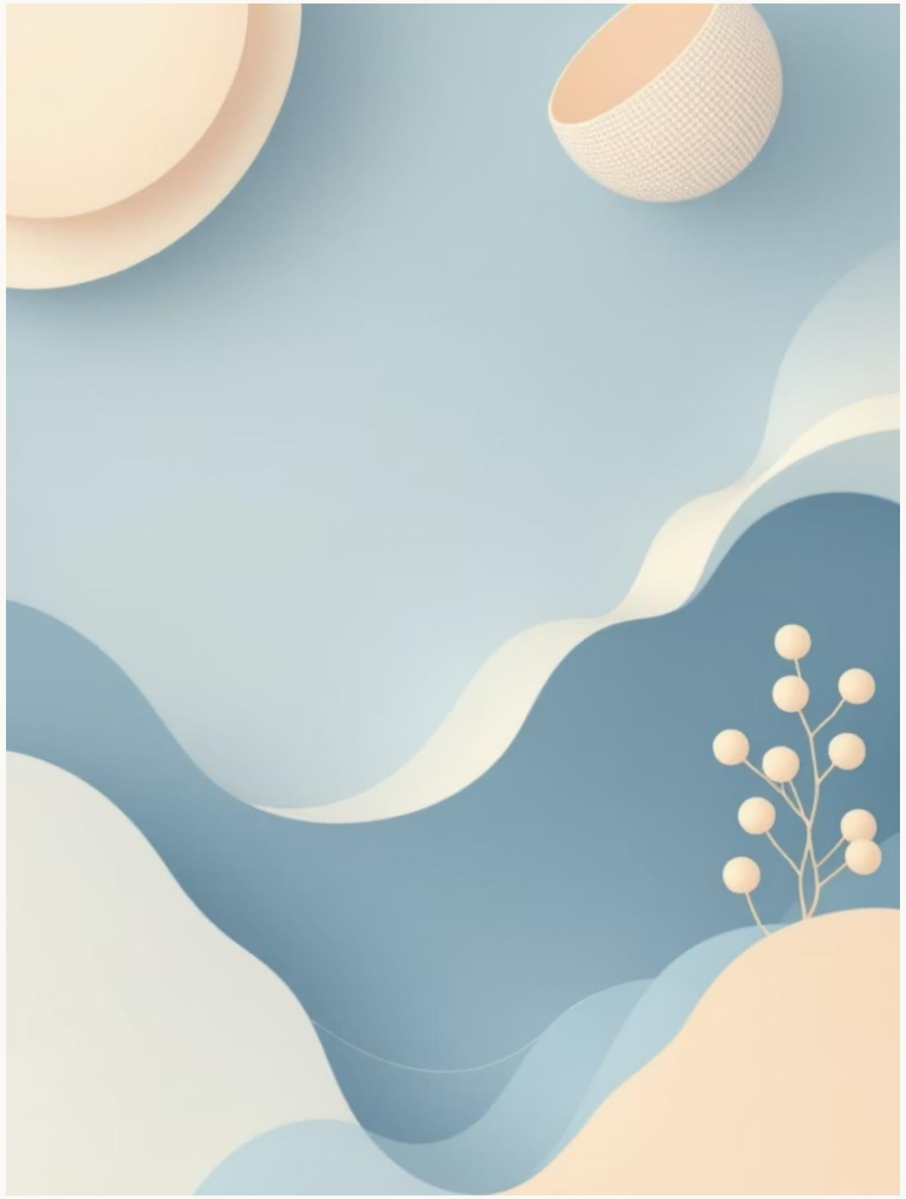
1. Randomly **drop** (set to 0) a fraction p of neurons
2. Forward pass + backprop with the dropped network
3. Update weights normally

Inference rule

- Use the full network (no dropout)
- Many frameworks scale weights automatically ("inverted dropout")

Student interpretation

Dropout acts like training **many small networks** and averaging them.



How L1 and L2 Regularization Work

Algorithm 5: Early Stopping

When to use

- Any iterative training (GD/SGD, deep nets)
- You have validation data

Procedure

1. Split data → Train / Validation
2. Train for epochs
3. If validation loss stops improving for K epochs (patience), stop training
4. Keep the best model weights (lowest validation loss)

Student interpretation

Stop before the model starts memorizing noise.

Algorithm 6: Data Augmentation (for images/audio/text)

When to use

- Small datasets, especially in vision

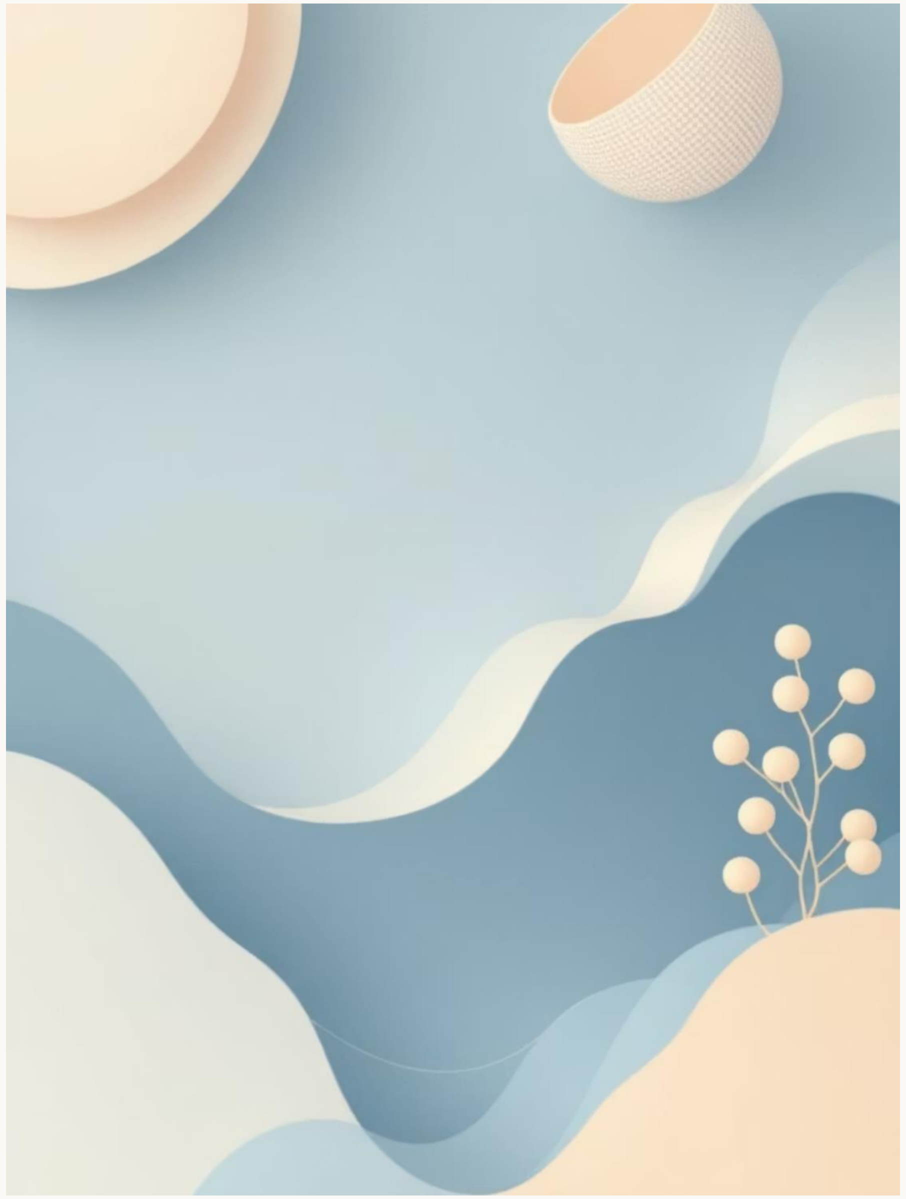
Procedure

Generate more training examples:

- image: flip, rotate, crop, noise
- audio: time shift, noise
- text: synonym replace (carefully)

Student interpretation

Makes the model robust by seeing “realistic variations”.



The L2 compromise in action

Scenario: A model has one parameter w. The original loss wants w=4. We add an L2 penalty where λ=1.

Original Loss: $J(w) = (w-4)^2$

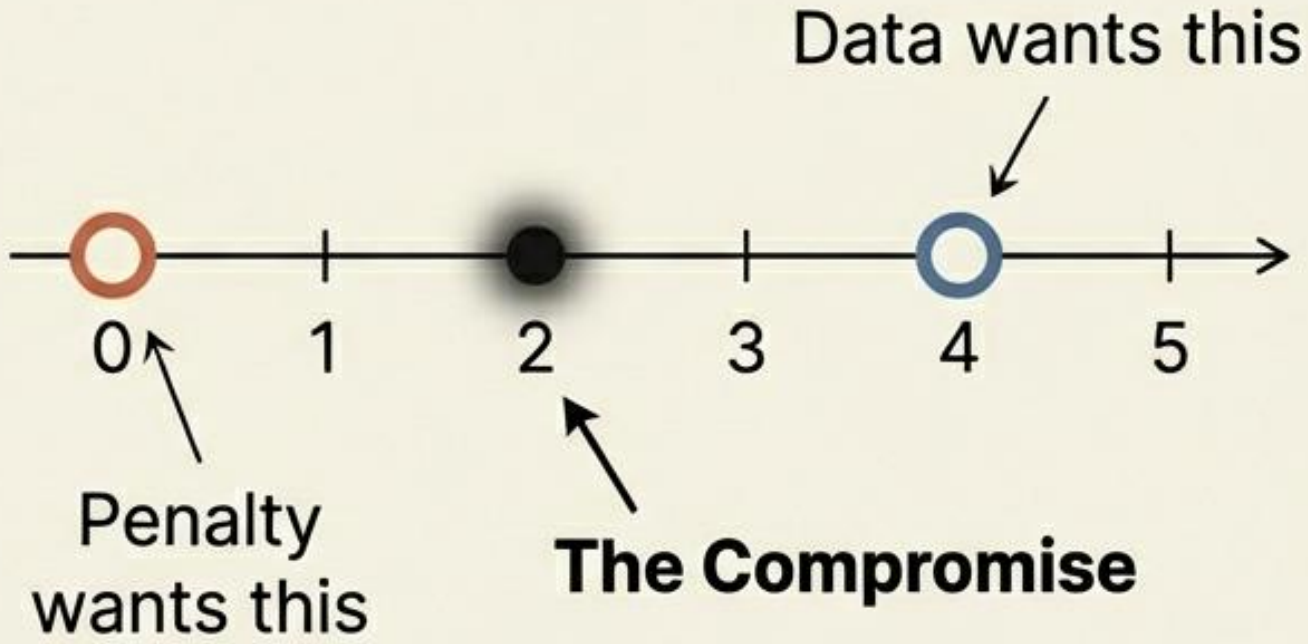
Regularized Loss: $J_{\text{reg}}(w) = (w-4)^2 + w^2$

Expanded: $J_{\text{reg}}(w) = 2w^2 - 8w + 16$

Derivative: $\frac{dJ_{\text{reg}}}{dw} \,/\, dw = 4w - 8 = 0$

Result: w = 2

Proof: $J_{reg}(2) = 8$, which is less than $J_{reg}(4) = 16$. w=2 is the mathematically superior choice.



Simple Solved Example

Solved Example: L2 Regularization

Problem

Suppose our model has only **one parameter** w, and the original loss is:

$$\boldsymbol{J(w) = (w - 4)^2}$$

Without regularization, the minimum is clearly at: $w = 4$

Now add **L2 regularization**: $\boldsymbol{J_{reg}(w) = (w - 4)^2 + \lambda w^2}$
Let: $\lambda = 1$, So the new loss becomes: $\boldsymbol{J_{reg}(w) = (w - 4)^2 + w^2}$
Step 1: Expand the expression

$$\boldsymbol{J_{reg}(w) = 2w^2 - 8w + 16}$$

Step 2: Differentiate $\frac{dJreg}{dw} = 4w - 8$
Step 3: Set derivative equal to zero $4w - 8 = 0, \ \ w = 2$

So the minimum occurs at: $\boxed{\boldsymbol{w = 2}}$

Compare with no regularization

Without regularization $J(w) = (w - 4)^2$

Minimum at: $\boxed{w = 4}$

With L2 regularization $J_{reg}(w) = (w - 4)^2 + w^2$
Minimum at: $\boxed{w = 2}$

So, the Main idea is Regularization **pulls the parameter toward zero**.

- without regularization: $w = 4$
- with regularization: $w = 2$

Very simple interpretation

The original loss wants $w$to become 4.
The regularization term wants w to become small.
The final answer is a compromise between fitting the data and keeping the weight small.

$$\boldsymbol{J_{reg}(2) < J_{reg}(4)}$$

which confirms that $\boldsymbol{w = 2}$ is better for the regularized problem.

Simple Solved Example

Solved Example: L1 Regularization

Problem

Suppose the original loss is:

$$\boldsymbol{J(w) = (w - 4)^2}$$

Without regularization, the minimum is: $w = 4$

Now add **L1 regularization**:

$$\boldsymbol{J_{reg}(w) = (w - 4)^2 + \lambda \mid w \mid}$$

Let: $\lambda = 2$

So the new loss becomes:

$$\boldsymbol{J_{reg}(w) = (w - 4)^2 + 2 \mid w \mid}$$

Case 1: $\boldsymbol{w \geq 0}$

If $\boldsymbol{w \geq 0}$, then: $\mid w \mid = w$

So: $\boldsymbol{J_{reg}(w) = (w - 4)^2 + 2w}$

Expand: $(w - 4)^2 = w^2 - 8w + 16$

Thus: $J_{reg}(w) = w^2 - 6w + 16$

Differentiate: $\boldsymbol{\frac{dJ_{reg}}{dw} = 2w - 6}$

Set equal to zero: $2w - 6 = 0, w = 3$

Since $3 \geq 0$, this solution is valid.

Case 2: $\boldsymbol{w < 0}$

If $w < 0$, then: $\mid w \mid = -w$

So: $J_{reg}(w) = (w - 4)^2 - 2w$

Expand: $J_{reg}(w) = w^2 - 10w + 16$

Differentiate: $\frac{dJ_{reg}}{dw} = 2w - 10$

Set equal to zero: $2w - 10 = 0,\ w = 5$

But this does **not** satisfy $w < 0$, so this case gives **no valid solution**.

Main idea for students

L1 regularization also **pushes the weight toward zero**.

- without regularization: $w = 4$
- with L1 regularization: $w = 3$

So L1 makes the model simpler.

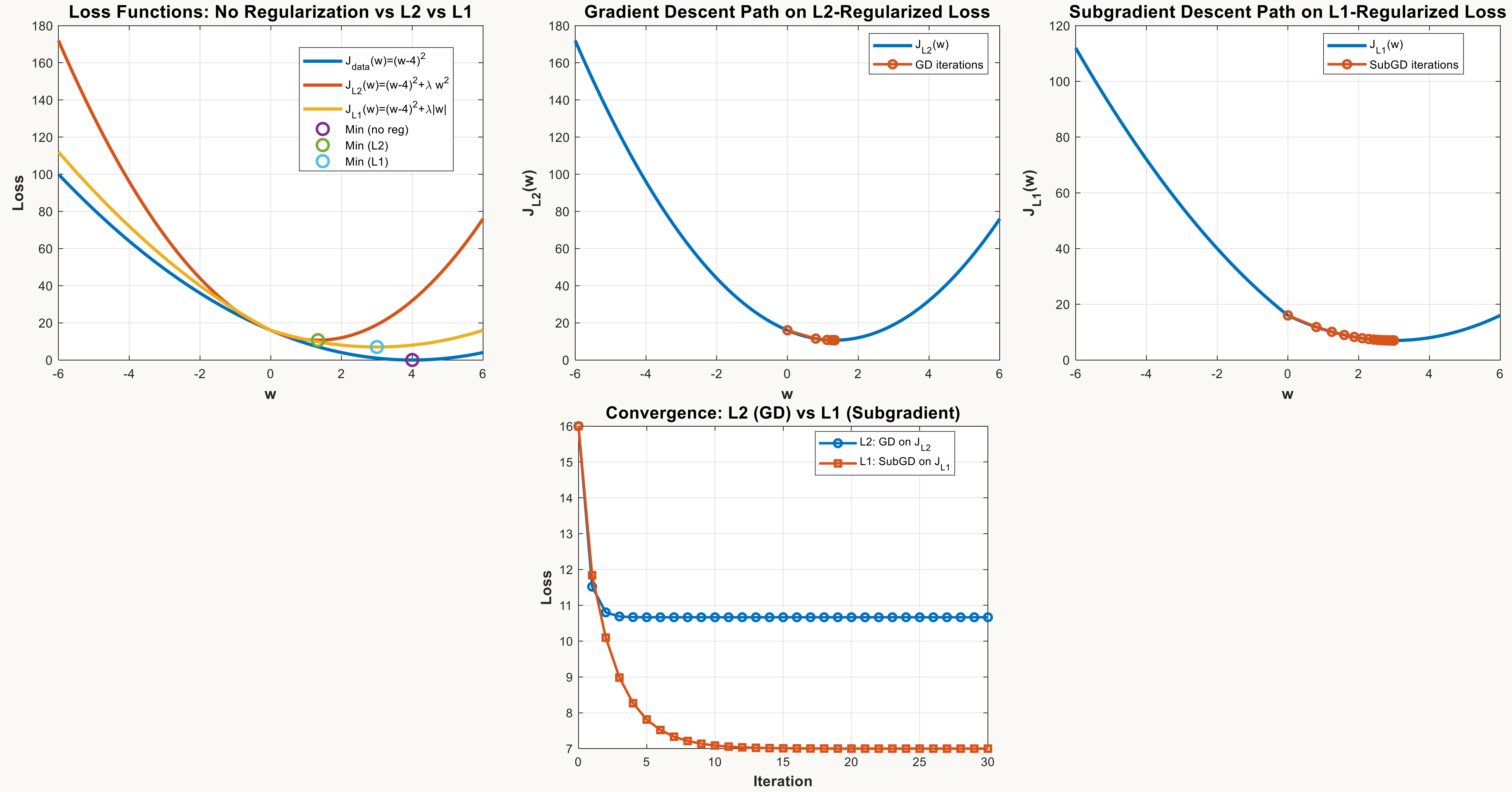
Why L1 is special

Unlike L2, L1 can push some weights **exactly to zero** in many problems.

That is why L1 is useful for:

- **feature selection**
- **sparse models**

Simple Solved Example



Summary: Choosing and Tuning Regularization

Reduces Overfitting

Constrains model complexity to improve robustness on unseen data

L1 for Sparsity

Ideal for feature selection—eliminates irrelevant features automatically

L2 for Stability

Smooth shrinkage of all weights—maintains all features while reducing magnitude

Dropout & Early Stopping

Essential for neural networks—complements weight regularization

Tuning λ: Use cross-validation to find the optimal regularization strength that minimizes validation error without underfitting.

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THANK you

Constrained Optimization

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Constrained Optimization

Types of constraints

Equality constraints (must be exactly satisfied):

$$g_i(x) = 0$$

Inequality constraints (must stay within a limit):

$$h_j(x) \leq 0$$

Box constraints (common in ML):

$$l \leq x \leq u$$

Feasible region

The **feasible set** is the set of points that satisfy all constraints:

$$\mathcal{F} = \{x \mid g_i(x) = 0,\, h_j(x) \leq 0\}$$

Optimization becomes:

$$x^* = \arg\min_{x \in \mathcal{F}} f(x)$$

HISTORICAL NOTES

Joseph-Louis Lagrange (1736–1813) Mathematician who developed many fundamental techniques in the calculus of variations, including the method that bears his name. Lagrange was largely self-taught, but quickly attracted the attention of the great mathematician Leonhard Euler. At age 19, Lagrange was appointed Professor of Mathematics at the Royal Artillery School in his native Turin. Over a long and outstanding career, Lagrange made contributions to probability, differential equations and fluid mechanics, for which he introduced what is now known as the Lagrangian function.

Constrained Optimization

Equality constraints: Lagrange multipliers

To solve:

$$\min_{x} f(x) \text{ s.t. } g(x) = 0$$

Define the **Lagrangian**:

$$\mathcal{L}(x,\lambda) = f(x) + \lambda^T g(x)$$

At optimum (first-order condition):

$$\nabla_x \mathcal{L}(x,\lambda) = 0, g(x) = 0$$

Interpretation: λmeasures how "costly" the constraint is (shadow price).

- substituting x_n into the objective function yields:

$$y = f[x_1, x_2, \ldots, x_{n-1}, B(x_1, x_2, \ldots, x_{n-1})]$$

- The necessary condition for an extreme point to be found at **X*** is given by:

$$\left.\frac{\partial y}{\partial x_j}\right|_{\mathbf{X}^*} = 0 \quad \forall j = 1, \ldots, n-1$$

Lagrange Multipliers

- Consider the problem of minimizing a continuous and differentiable function $y=f(\mathbf{X})$ subject to one equality constraint in the form $g(\mathbf{X})=0$
- If a variable x_n in the constraint is selected and expressed in terms of the remaining $(n-1)$ variables such that:

$$x_n = B(x_1, x_2, \ldots, x_{n-1})$$

- Applying the chain rule:

$$\left.\frac{\partial y}{\partial x_j}\right|_{\mathbf{X}^*} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{X}^*} + \left.\frac{\partial f}{\partial B}\cdot\frac{\partial B}{\partial x_j}\right|_{\mathbf{X}^*} = 0$$

$$\forall j = 1, \ldots, n-1$$

$$\therefore \left.\frac{\partial y}{\partial x_j}\right|_{\mathbf{X}^*} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{X}^*} + \left.\frac{\partial f}{\partial x_n}\cdot\frac{\partial B}{\partial x_j}\right|_{\mathbf{X}^*} = 0$$

$$\forall j = 1, \ldots, n-1$$

Constrained Optimization

- Differentiating the constraint:

$$\frac{\partial g}{\partial x_j} + \frac{\partial g}{\partial B} \cdot \frac{\partial B}{\partial x_j} = 0$$

$$\forall j = 1, \ldots, n-1$$

$$\frac{\partial g}{\partial x_j} + \frac{\partial g}{\partial x_n} \cdot \frac{\partial B}{\partial x_j} = 0$$

$$\forall j = 1, \ldots, n-1$$

$$\therefore \frac{\partial B}{\partial x_j} = -\frac{\partial g}{\partial x_j} \Big/ \frac{\partial g}{\partial x_n} \qquad j = 1, \ldots, n-1$$

- given that:

$$\frac{\partial g}{\partial x_n} \neq 0$$

- Thus, the necessary condition for optimality becomes:

$$\left.\frac{\partial y}{\partial x_j}\right|_{\mathbf{X}^*} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{X}^*} - \left[\frac{\partial f}{\partial x_n} \cdot \frac{\partial g}{\partial x_j} \Big/ \frac{\partial g}{\partial x_n}\right]_{\mathbf{X}^*} = 0$$

- Let:

$$\lambda = -\frac{\partial f}{\partial x_n} \Big/ \frac{\partial g}{\partial x_n}$$

- so that, the necessary condition for $\mathbf{X}^*$ to be an optimal point is:

$$\boxed{\left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{X}^*} + \lambda \left.\frac{\partial g}{\partial x_j}\right|_{\mathbf{X}^*} = 0, \quad (1)}$$

$$, j = 1, \ldots, n-1$$

- Consider a function $L(\mathbf{X})$:

$$L(\mathbf{X}) = f(\mathbf{X}) + \lambda g(\mathbf{X})$$

- The necessary condition for the such function to have an extreme point at $\mathbf{X}^*$ is:

$$\left.\frac{\partial L}{\partial x_j}\right|_{\mathbf{X}^*} = \left.\frac{\partial f}{\partial x_j}\right|_{\mathbf{X}^*} + \lambda \left.\frac{\partial g}{\partial x_j}\right|_{\mathbf{X}^*} = 0 \qquad (2)$$

$$\left.\frac{\partial L}{\partial \lambda}\right|_{\mathbf{X}^*} = g(\mathbf{X}) = 0$$

- Equation (2) is the same as equation (1) which is the necessary condition.
- The function $L(\mathbf{X})$ is known as the Lagrangian function.
- The necessary condition for an extreme point of the lagrangian function is equivalent to the necessary condition for an extreme point of the corresponding equality constrained problem

- An extreme point $\mathbf{X}^*$ must satisfy both the necessary condition and the constraint $g(\mathbf{X}^*)=0$
- These constitute $n+1$ equations in $n+1$ variables, which are the elements of vector $\mathbf{X}^*$ in addition to λ.
- The variable λ is known as the Lagrange multiplier and it represents the rate of change of the objective f with respect to the constraint g

Example(1): Optimization problem with one equality constraint

- Maximize:

$$f(x_1,x_2) = 3x_1^2 + x_2^2 + 2x_1x_2 + 6x_1 + 2x_2$$

- Subject to:

$$2x_1 - x_2 = 4$$

Solution

- Form the Lagrangian function:

$$L(x_1,x_2,\lambda) = 3x_1^2 + x_2^2 + 2x_1x_2 + 6x_1 + 2x_2 + \lambda(2x_1 - x_2 - 4)$$

- The necessary conditions are:

$$\partial L/\partial x_1 = 6x_1 + 2x_2 + 6 - 2\lambda = 0$$

$$\partial L/\partial x_2 = 2x_2 + 2x_1 + 2 + \lambda = 0$$

$$\partial L/\partial \lambda = 2x_1 - x_2 - 4 = 0$$

- Solving the 3 equations:

$$x_1^* = \frac{7}{11}$$

$$x_2^* = \frac{-30}{11}$$

$$\lambda = \frac{24}{11}$$

Example(2):Optimization problem with one equality constraint

- Minimize:

$$f(x,y) = kx^{-1}y^{-2}$$

- Subject to:

$$g(x,y) = x^2 + y^2 - a^2$$

Solution

- The Lagrange function:

$$L(x,y,\lambda) = f(x,y) + \lambda g(x,y)$$

$$= kx^{-1}y^{-2} + \lambda\left(x^2 + y^2 - a^2\right)$$

- The necessary condition for the minimum of $f(x)$ is:

$$\frac{\partial L}{\partial x} = -kx^{-2}y^{-2} + 2x\lambda = 0$$

$$\frac{\partial L}{\partial y} = -2kx^{-1}y^{-3} + 2y\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - a^2 = 0$$

- Solving equations 1,2,3:

$$2\lambda = \frac{k}{x^3y^2} = \frac{2k}{xy^4}$$

$$\therefore x^* = \left(1/\sqrt{2}\right)y^*$$

$$\therefore x^* = \frac{a}{\sqrt{3}}$$

$$y^* = \sqrt{\frac{2}{3}}a$$

Necessary Conditions for a Problem with *n* equality constraints

- Minimize:

$$f(x_1, x_2, \ldots, x_n)$$

- Subject to:

$$g_i(x_1, x_2, \ldots, x_n) = 0, i = 1,2,\ldots,m$$

- the Lagrange function:

$$L(\mathbf{X}, \boldsymbol{\lambda}) = f(\mathbf{X}) + \sum_{j=1}^{m} \lambda_j g_j(\mathbf{X})$$

- The vector λ is the vector of Lagrange multipliers $\boldsymbol{\lambda} = [\lambda_1, \lambda_2, \ldots, \lambda_n]$

Necessary Conditions for a Problem with *n* equality constraints

- the necessary condition for the Lagrange function to have an extreme point at **X*** is:

$$\left.\frac{\partial L}{\partial x_i}\right|_{\mathbf{X}^*} = \left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{X}^*} + \sum_{j=1}^{m} \lambda_j \left.\frac{\partial g_j}{\partial x_i}\right|_{\mathbf{X}^*} = 0, i = 1,2,\ldots,n$$

$$\left.\frac{\partial L}{\partial \lambda_j}\right|_{\mathbf{X}^*} = g_j(\mathbf{X}) = 0, j = 1,2,\ldots,m$$

Example(3): on the necessary conditions for the general problem

- Minimize:

$$f(\mathbf{X}) = x_1^2 + x_2^2 + x_3^2$$

- Subject to:

$$g_1(\mathbf{X}) = x_1 + x_2 + 3x_3 - 2 = 0$$

$$g_2(\mathbf{X}) = 5x_1 + 2x_2 + x_3 - 5 = 0$$

Solution

- Formulate the Lagrangian function:

$$L(\mathbf{X}, \lambda) = x_1^2 + x_2^2 + x_3^2 + \lambda_1(x_1 + x_2 + 3x_3 - 2) + \lambda_2(5x_1 + 2x_2 + x_3 - 5)$$

- The necessary conditions are:

$$\partial L/\partial x_1 = 2x_1 + \lambda_1 + 5\lambda_2 = 0$$

$$\partial L/\partial x_2 = 2x_2 + \lambda_1 + 2\lambda_2 = 0$$

$$\partial L/\partial x_3 = 2x_3 + 3\lambda_1 + \lambda_2 = 0$$

$$\partial L/\partial \lambda_1 = x_1 + x_2 + 3x_3 - 2 = 0$$

$$\partial L/\partial \lambda_2 = 5x_1 + 2x_2 + x_3 - 5 = 0$$

- Solving equations 1 thru 5, gives:

$$\mathbf{X}^* = [x_1 \quad x_2 \quad x_3] = [0.81 \quad 0.35 \quad 0.28]$$

- and:

$$\boldsymbol{\lambda} = [\lambda_1 \quad \lambda_2] = [0.0867 \quad 0.3067]$$

Sufficient Condition

- In all the examples solved before, there is no guarantee that any solution obtained is a maximum or a minimum.
- The solution obtained must be verified to be an extreme point by applying the sufficient condition

- A sufficient condition for $L(\mathbf{X})$ to have a relative minimum at $\mathbf{X}^*$ is that the quadratic, Q, defined by:

$$Q = \sum_{i=1}^{n} \sum_{j=1}^{n} \frac{\partial^2 L}{\partial x_i \partial x_j} dx_i dx_j$$

Evaluated at $\mathbf{X} = \mathbf{X}^*$ must be positive definite for all values of $d\mathbf{X}$ for which the constraints are satisfied.

Notes on the Sufficient condition

- If Q is negative for all choices of the admissible variations $d\mathbf{X}$,, then $\mathbf{X}^*$ will be a constrained maximum of $f(\mathbf{X})$.
- It has been shown that the necessary conditions for the quadratic form Q, to be positive definite for all admissible variations $d\mathbf{X}$ is that each root of the polynomial z_i defined by the following determinant equation be positive.

Notes on the Sufficient condition

$$\begin{vmatrix}
L_{11}-z & L_{12} & L_{13} & \cdots & L_{1n} & g_{11} & g_{21} & \cdots & g_{m1} \\
L_{21} & L_{22}-z & L_{23} & \cdots & L_{2n} & g_{12} & g_{22} & \cdots & g_{m2} \\
\vdots & & & & & & & & \\
L_{n1} & L_{n2} & L_{n3} & \cdots & L_{nn}-z & g_{1n} & g_{2n} & \cdots & g_{mn} \\
g_{11} & g_{12} & g_{13} & \cdots & g_{1n} & 0 & 0 & \cdots & 0 \\
g_{21} & g_{22} & g_{23} & \cdots & g_{2n} & 0 & 0 & \cdots & 0 \\
\vdots & & & & & & & & \\
g_{m1} & g_{m2} & g_{m3} & \cdots & g_{mn} & 0 & 0 & \cdots & 0
\end{vmatrix} = 0$$

Where: $L_{ij} = \left.\frac{\partial^2 L}{\partial x_i \partial x_j}\right|_{(\mathbf{X}^*, \lambda^*)}$ $\quad g_{ij} = \left.\frac{\partial g_i}{\partial x_j}\right|_{\mathbf{X}^*}$

Example 3

A closed cylindrical tin (with **top and bottom**) is made from sheet metal. The total surface area is fixed at

$$\boldsymbol{A_0 = 24\pi} \text{ (units}^2\text{)}.$$

Let the cylinder have radius $r$and height h.

(a) Formulate the optimization problem to **maximize the volume** subject to the surface-area constraint.

(b) Use the **Lagrange multiplier method** to find the optimal dimensions $\left(r^{\backslash *}, h^{\backslash *}\right)$.

(c) Compute the maximum volume $V_{\max}$.

Solution

(a) Formulation

Volume: $V(r,h) = \pi r^2 h$

Surface area (top + bottom + lateral):

$$A(r,h) = 2\pi r^2 + 2\pi rh = 24\pi$$

Optimization problem:

$$\max_{r,h} \ \pi r^2 h \text{s.t.} 2\pi r^2 + 2\pi rh - 24\pi = 0$$

(b) Lagrange multiplier method

Lagrangian: $\mathcal{L}(r, h, \lambda) = \pi r^2 h + \lambda(2\pi r^2 + 2\pi r h - 24\pi)$

First-order conditions:

1) $\boldsymbol{\frac{\partial \mathcal{L}}{\partial r} = 0}$ $\quad \frac{\partial \mathcal{L}}{\partial r} = 2\pi r h + \lambda(4\pi r + 2\pi h) = 0$

2) $\boldsymbol{\frac{\partial \mathcal{L}}{\partial h} = 0}$ $\quad \frac{\partial \mathcal{L}}{\partial h} = \pi r^2 + \lambda(2\pi r) = 0$

Divide by $\pi r \neq 0$: $\quad r + 2\lambda = 0 \Rightarrow \lambda = -\frac{r}{2}$

Substitute into the first equation: $\quad 2\pi r h + \left(\to \frac{r}{2} \right)(4\pi r + 2\pi h) = 0$

$$2\pi r h - 2\pi r^2 - \pi r h = 0 \Rightarrow \pi r h = 2\pi r^2 \Rightarrow h = 2r$$

3) Constraint equation

$$2\pi r^2 + 2\pi r h = 24\pi$$

Substitute $h = 2r$:

$$2\pi r^2 + 2\pi r(2r) = 24\pi \Rightarrow 2\pi r^2 + 4\pi r^2 = 24\pi \Rightarrow 6\pi r^2 = 24\pi \Rightarrow r^2 = 4 \Rightarrow r = 2$$

$$h = 2r = 4$$

So: $\boxed{r^{\backslash *} = 2, h^{\backslash *} = 4}$ and $\lambda^{\backslash *} = -\frac{r}{2} = -1$

(c) Maximum volume

$$V_{\max} = \pi r^{*2} h^{\backslash *} = \pi(2^2)(4) = 16\pi$$

$$\boxed{V_{\max} = 16\pi}$$

Example4

Use the method of **Lagrange multipliers** (the relationship $\boldsymbol{\nabla f = \lambda \nabla g}$) to find the point on the line

$$y = 3 - 2x$$

that is **closest to the origin**.

(a) Formulate the problem as minimizing a function $f(x, y)$ subject to a constraint $g(x, y) = 0$.
(b) Compute ∇f and ∇g, then solve $\nabla f = \lambda \nabla g$ with the constraint.
(c) State the closest point and the minimum distance to the origin.

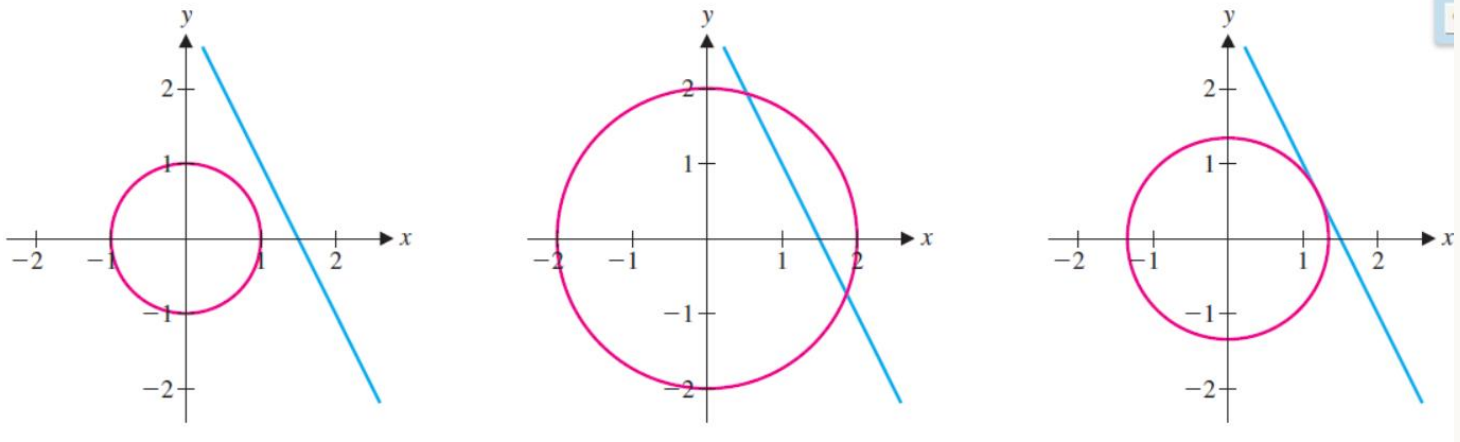

FIGURE 12.50b $y = 3 - 2x$ and the circle of radius 1 centered at (0, 0)

FIGURE 12.50c $y = 3 - 2x$ and the circle of radius 2 centered at (0, 0)

FIGURE 12.50d $y = 3 - 2x$ and a circle tangent to the line

(a) Formulation

Minimize the squared distance to the origin: $f(x, y) = x^2 + y^2$
Constraint (line written as $g(x, y) = 0$):

$$y = 3 - 2x \;\Rightarrow\; 2x + y - 3 = 0$$
$$g(x, y) = 2x + y - 3 = 0$$

(b) Lagrange multipliers

Compute gradients:

$$\nabla f = (2x,\ 2y), \nabla g = (2,\ 1)$$

Set: $\nabla f = \lambda \nabla g \Rightarrow (2x, 2y) = \lambda(2,1)$

So: $2x = 2\lambda \Rightarrow x = \lambda, 2y = \lambda \Rightarrow \lambda = 2y \Rightarrow x = 2y$

Use the constraint $2x + y - 3 = 0$. Substitute $x = 2y$:

$$2(2y) + y - 3 = 0 \Rightarrow 5y = 3 \Rightarrow y = \frac{3}{5}$$

Then: $x = 2y = \frac{6}{5}$, Closest point: $\boxed{\left(\frac{6}{5}, \frac{3}{5}\right)}$

(c) Minimum distance

Minimum squared distance: $f_{\min} = x^2 + y^2 = \left(\frac{6}{5}\right)^2 + \left(\frac{3}{5}\right)^2 = \frac{36}{25} + \frac{9}{25} = \frac{45}{25} = \frac{9}{5}$

So the minimum distance is: $d_{\min} = \sqrt{f_{\min}} = \sqrt{\frac{9}{5}} = \frac{3}{\sqrt{5}}$, $\boxed{d_{\min} = \frac{3}{\sqrt{5}}}$

EXAMPLE 8.3 Optimization with an Inequality Constraint

Suppose that the temperature of a metal plate is given by $T(x, y) = x^2 + 2x + y^2$, for points (x, y) on the elliptical plate defined by $x^2 + 4y^2 \leq 24$. Find the maximum and minimum temperatures on the plate.

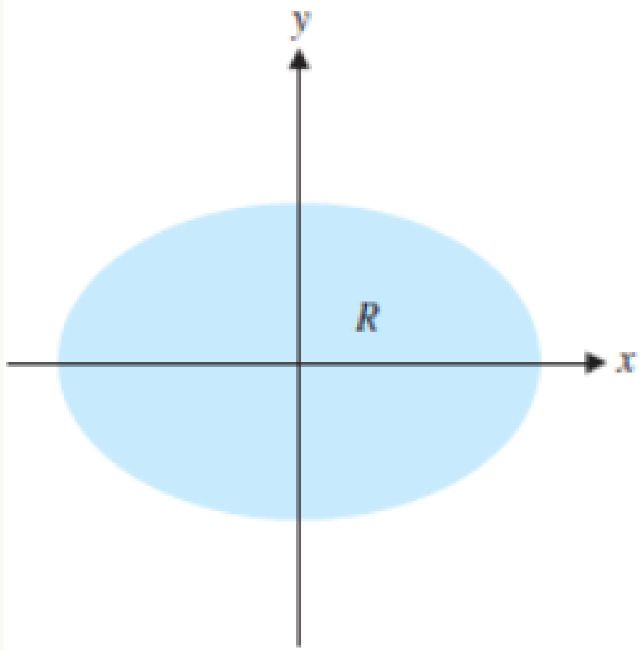

FIGURE 12.52
A metal plate

1. Interior Critical Points

Inside the ellipse, we look for points where the gradient of T is zero:

$$\nabla T = (2x + 2,\ 2y) = (0, 0) \quad \Rightarrow \quad 2x + 2 = 0,\ 2y = 0.$$

Thus $x = -1,\ y = 0$. Check that this point satisfies the inequality:

$$(-1)^2 + 4(0)^2 = 1 \leq 24,$$

so it is inside. The temperature there is

$$T(-1, 0) = (-1)^2 + 2(-1) + 0 = 1 - 2 = -1.$$

2. Boundary: $x^2 + 4y^2 = 24$

Use Lagrange multipliers with constraint $g(x, y) = x^2 + 4y^2 - 24 = 0$. Set $\nabla T = \lambda \nabla g$:

$$\begin{cases} 2x + 2 = \lambda(2x) \quad \Rightarrow \quad x + 1 = \lambda x & (1) \\ 2y = \lambda(8y) \quad \Rightarrow \quad 2y = 8\lambda y & (2) \end{cases}$$

We consider two cases.

Case 1: $y = 0$

Then the constraint gives $x^2 = 24$ so $x = \pm 2\sqrt{6}$. Equation (2) is automatically satisfied. For each x, equation (1) determines λ (not needed further). Compute temperatures:

- At $(2\sqrt{6}, 0)$: $T = (2\sqrt{6})^2 + 2(2\sqrt{6}) + 0 = 24 + 4\sqrt{6}$.
- At $(-2\sqrt{6}, 0)$: $T = 24 - 4\sqrt{6}$.

Numerically, $4\sqrt{6} \approx 9.798$, so these are about 33.798 and 14.202.

Case 2: $y \neq 0$

From equation (2), divide by $2y$ (since $y \neq 0$):

$$1 = 4\lambda \quad \Rightarrow \quad \lambda = \frac{1}{4}.$$

Substitute into (1):

$$x + 1 = \frac{1}{4}x \quad \Rightarrow \quad 4x + 4 = x \quad \Rightarrow \quad 3x = -4 \quad \Rightarrow \quad x = -\frac{4}{3}.$$

Now use the constraint:

$$\left(-\frac{4}{3}\right)^2 + 4y^2 = 24 \quad \Rightarrow \quad \frac{16}{9} + 4y^2 = 24 \quad \Rightarrow \quad 4y^2 = 24 - \frac{16}{9} = \frac{216 - 16}{9} = \frac{200}{9}$$

Thus

$$y^2 = \frac{200}{36} = \frac{50}{9} \quad \Rightarrow \quad y = \pm\frac{5\sqrt{2}}{3}.$$

The two points are $\left(-\frac{4}{3}, \frac{5\sqrt{2}}{3}\right)$ and $\left(-\frac{4}{3}, -\frac{5\sqrt{2}}{3}\right)$. Compute the temperature:

$$T = x^2 + 2x + y^2 = \frac{16}{9} - \frac{8}{3} + \frac{50}{9} = \frac{16 + 50}{9} - \frac{24}{9} = \frac{66 - 24}{9} = \frac{42}{9} = \frac{14}{3} \approx 4.667.$$

3. Compare Values

We have the following candidate temperatures:

- Interior: -1
- Boundary: $24 + 4\sqrt{6} \approx 33.798$
- Boundary: $24 - 4\sqrt{6} \approx 14.202$
- Boundary: $\frac{14}{3} \approx 4.667$

The largest is $24 + 4\sqrt{6}$ at $(2\sqrt{6}, 0)$; the smallest is -1 at $(-1, 0)$.

4. Conclusion

The maximum temperature on the plate is $24 + 4\sqrt{6}$ and the minimum temperature is -1.

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THANK you

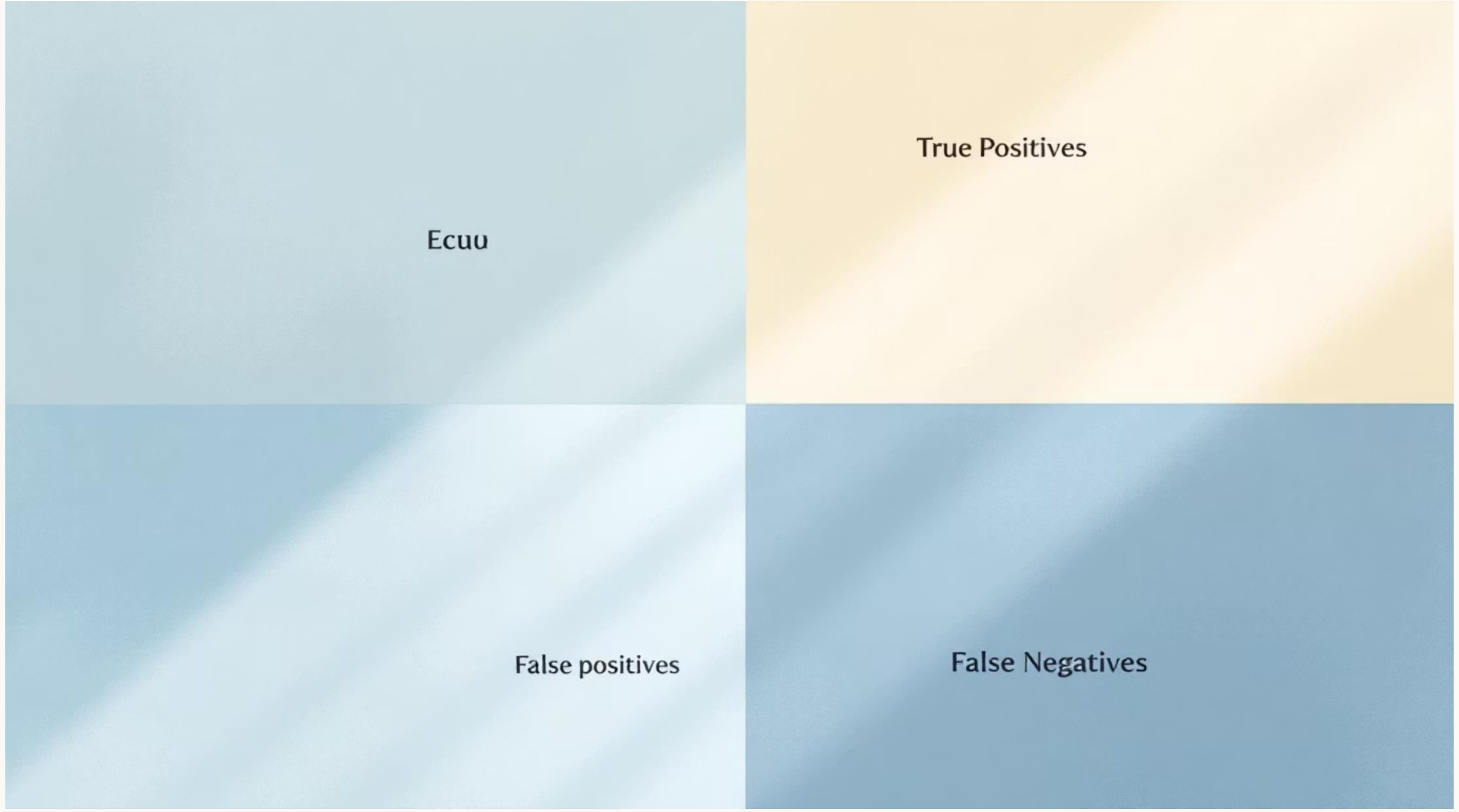
Model Evaluation Metrics

A. prof. Elsayed Mahdy

The Foundation: Understanding the Confusion Matrix

The confusion matrix is the cornerstone of classification evaluation. It breaks down predictions into four fundamental outcomes that reveal how well your model performs across different scenarios.

- **True Positives (TP)**: Correctly predicted positive cases
- **True Negatives (TN)**: Correctly predicted negative cases
- **False Positives (FP)**: Negative cases incorrectly predicted as positive
- **False Negatives (FN)**: Positive cases incorrectly predicted as negative



Visualizing predictions against actual labels helps identify where your model succeeds and where it struggles, providing insights for improvement.

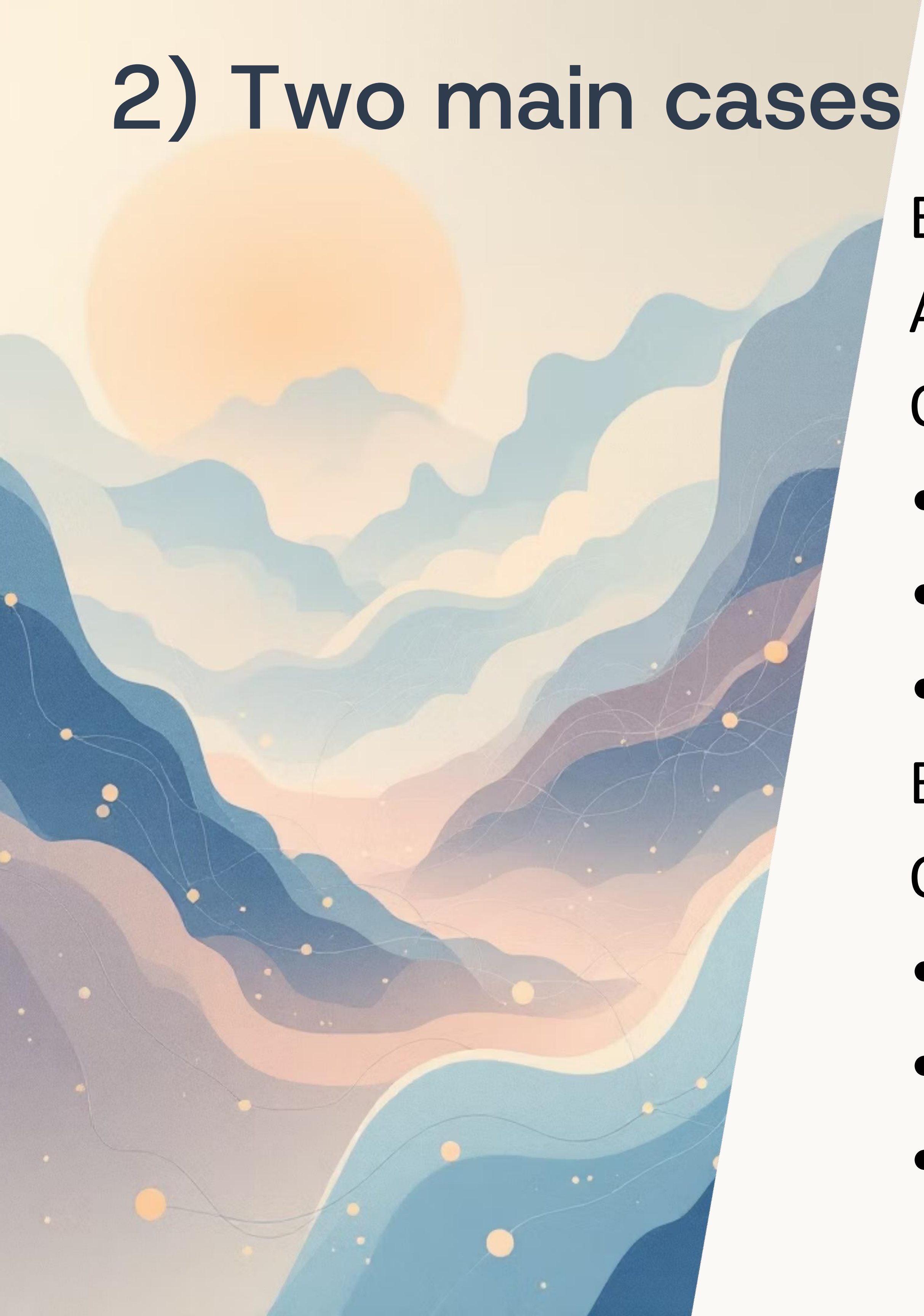
1) Why do we need evaluation metrics?

After training a model, we must answer:

- **How good is the model?**
- **Is it making correct predictions?**
- **Can we trust it on new data?**

Model evaluation metrics help us measure the performance of a machine learning model.

2) Two main cases



Evaluation metrics depend on the type of problem:

A) Classification

Output is a category, such as:

- spam / not spam
- disease / no disease
- cat / dog

B) Regression

Output is a numerical value, such as:

- house price
- temperature
- radiation level

2) Two main cases

Evaluation metrics depend on the type of problem:

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Output is a numerical value, such as:

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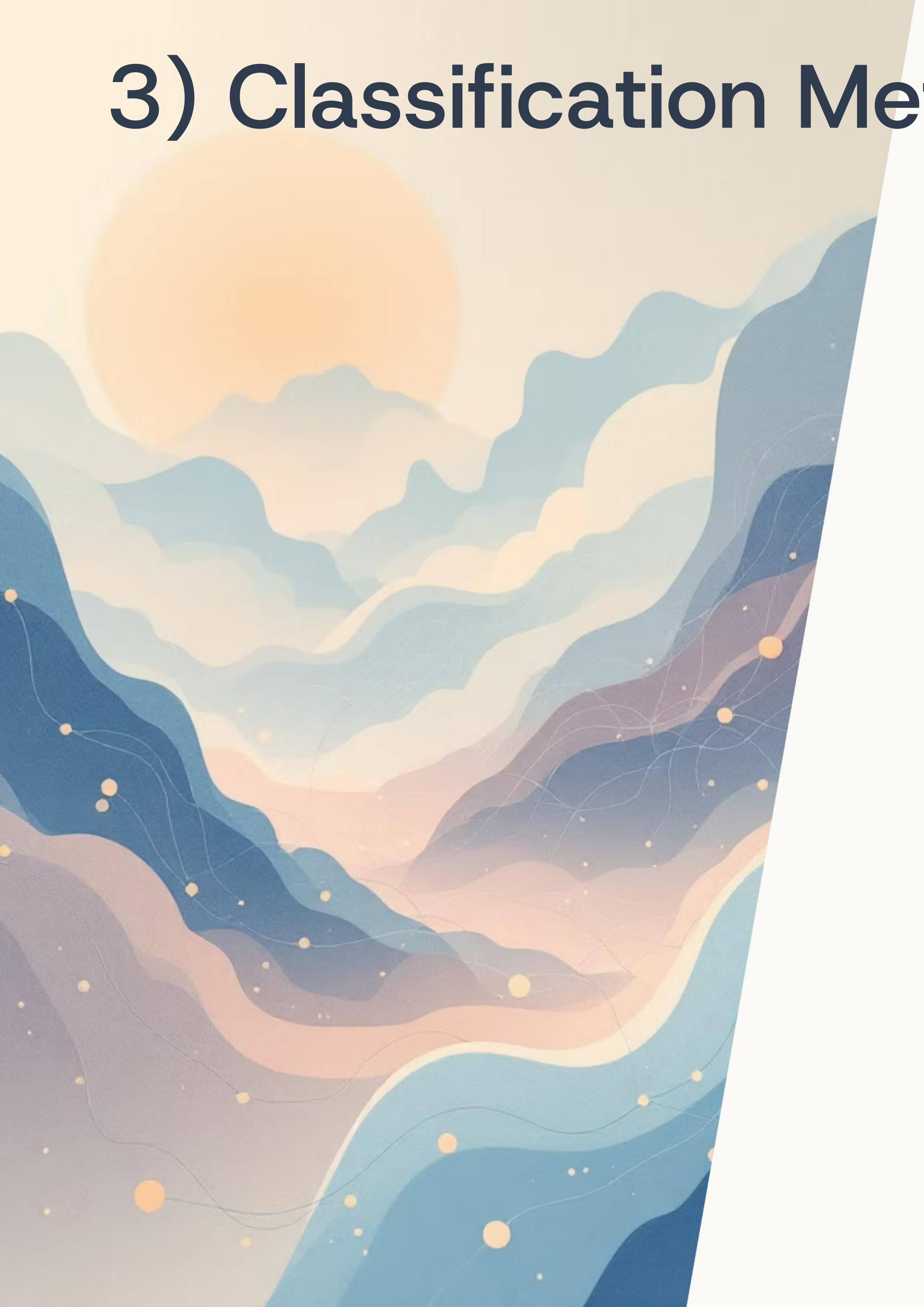
3) Classification Metrics



Classification usually starts with the **confusion matrix**.

Confusion Matrix

For binary classification:

- **TP** = True Positive
  Model predicts positive, and it is actually positive
- **TN** = True Negative
  Model predicts negative, and it is actually negative
- **FP** = False Positive
  Model predicts positive, but it is actually negative
- **FN** = False Negative
  Model predicts negative, but it is actually positive

Accuracy: The Intuitive but Misleading Metric

Accuracy tells us the overall fraction of correct predictions.

$$\textbf{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Meaning

Out of all predictions, how many were correct?

Limitation

Accuracy can be misleading in **imbalanced datasets**.

Example:
If 95% of samples are negative, a model that predicts all negative gets 95% accuracy, but it is useless.

Precision: Trusting Positive Predictions

Precision tells us:

$$\textbf{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FP}}$$

Meaning

Out of all predicted positives, how many are truly positive?

Important when:

False positives are costly.

Example:

- email spam detection
- fraud detection

Recall (Sensitivity): Catching All Positives

Recall tells us:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Meaning

Out of all actual positives, how many did the model correctly find?

Important when:

False negatives are dangerous.

Example:

- **disease diagnosis**
- **fire detection**
- **radiation anomaly detection**

F1-Score

F1-score balances precision and recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Meaning

A single score that combines precision and recall.

Important when:

Classes are imbalanced and both FP and FN matter.

Specificity

Specificity tells us:

$$\textbf{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN + FP}}$$

Meaning

Out of all actual negatives, how many did the model correctly identify?

This is useful in medical and safety applications.

ROC Curve and AUC
ROC Curve

Plots:

- True Positive Rate (Recall)
- versus False Positive Rate

AUC

Area Under the Curve

- **AUC = 1 → perfect model**
- **AUC = 0.5 → random guessing**

Meaning

Shows how well the classifier separates the classes.

Specificity

Specificity tells us:

$$\textbf{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN} + \boldsymbol{FP}}$$

Meaning

Out of all actual negatives, how many did the model correctly identify?

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ROC Curve and AUC

ROC Curve

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AUC

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Meaning

Shows how well the classifier separates the classes.

4) Regression Metrics: Regression models predict continuous values.

(1) MAE — Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |\, y_i - \hat{y}_i \,|$$

Meaning

Average absolute difference between actual and predicted values.

Advantage

Easy to understand.

(2) MSE — Mean Squared Error

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Meaning

Average squared error.

Advantage

Penalizes large errors more strongly.

(3) RMSE — Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

Meaning

Same idea as MSE, but in the original unit of the target.

Example:
If target is temperature in °C, RMSE is also in °C.

(4) R^2Score (Coefficient of Determination)

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Meaning

Measures how much variance in the target is explained by the model.

- $R^2 = 1 \rightarrow$ perfect fit
- $R^2 = 0 \rightarrow$ model is no better than predicting the mean
- $R^2 < 0 \rightarrow$ model is worse than mean prediction

5) When to use which metric?

For classification

- **Accuracy** → balanced classes
- **Precision** → reduce false positives
- **Recall** → reduce false negatives
- **F1-score** → balance between precision and recall
- **AUC** → overall classification quality

For regression

- **MAE** → simple average error
- **MSE / RMSE** → penalize large mistakes
- R^2 → overall goodness of fit

Question:

Given the following confusion matrix values for a binary classification model:

True Positives (TP) = 40

True Negatives (TN) = 50

False Positives (FP) = 5

False Negatives (FN) = 5

Calculate the **accuracy**, **precision**, **recall**, and **F1-score**. Express each result as a percentage.

$$\text{Accuracy} = \frac{40+50}{40+50+5+5} = \frac{90}{100} = 0.9$$

$$\boxed{\text{Accuracy} = 90\%}$$

$$\text{Precision} = \frac{40}{40+5} = \frac{40}{45} = 0.8889$$

$$\boxed{\text{Precision} \approx 88.89\%}$$

$$\text{Recall} = \frac{40}{40+5} = \frac{40}{45} = 0.8889$$

$$\boxed{\text{Recall} \approx 88.89\%}$$

$$F1 = 2 \cdot \frac{0.8889 \cdot 0.8889}{0.8889 + 0.8889} = 0.8889$$

$$\boxed{F1 \approx 88.89\%}$$

Very simple solved regression example

Suppose:

$$y = [3,5,7]$$
$$\hat{y} = [2.5,5.5,6]$$

Step 1: Errors

$$y - \hat{y} = [0.5, -0.5, 1]$$

MAE

$$MAE = \frac{|\,0.5\,|+|\,-0.5\,|+|\,1\,|}{3} = \frac{0.5+0.5+1}{3} = \frac{2}{3} = 0.667$$

$$\boxed{MAE = 0.667}$$

MSE

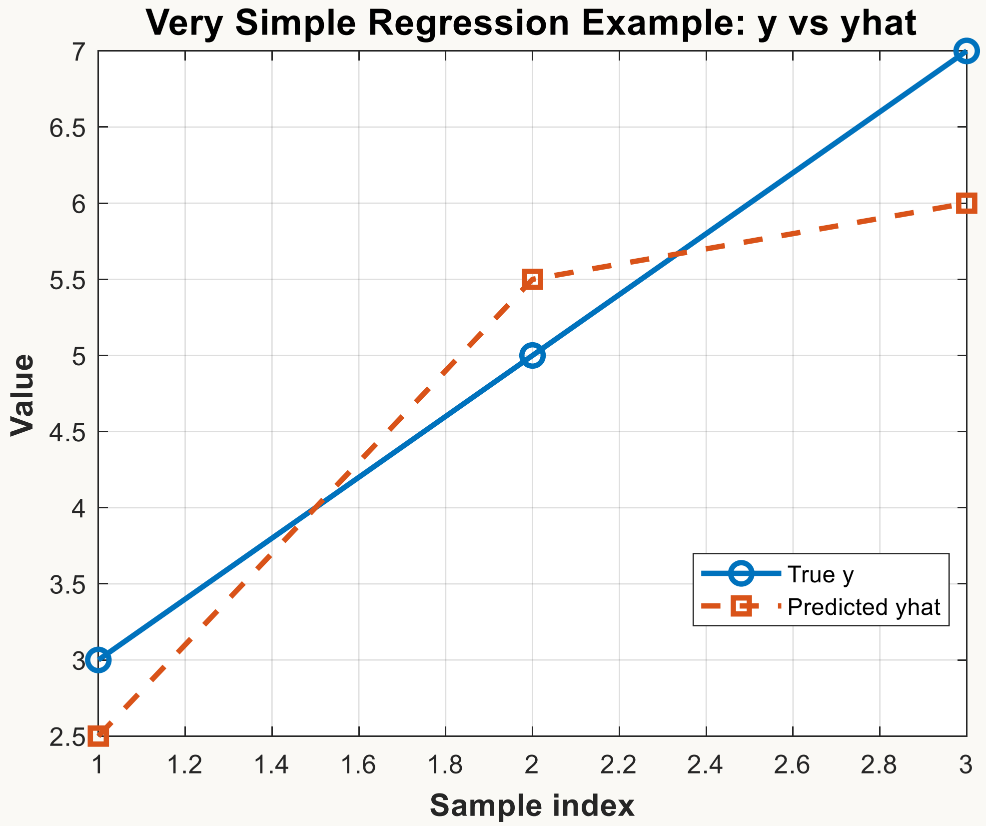
$$MSE = \frac{(0.5)^2 + (-0.5)^2 + (1)^2}{3}$$
$$= \frac{0.25+0.25+1}{3} = \frac{1.5}{3} = 0.5$$

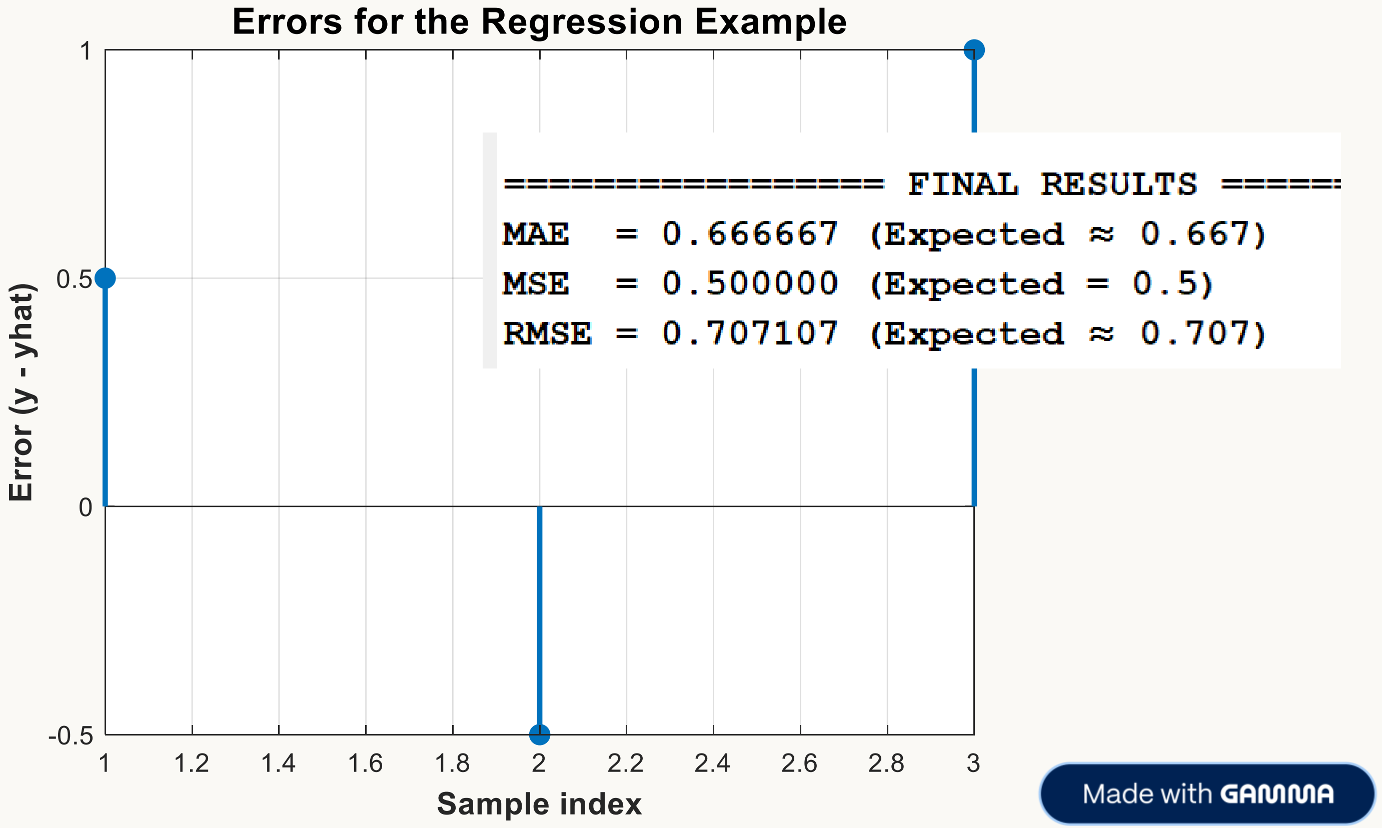
$$\boxed{MSE = 0.5}$$

RMSE

$$RMSE = \sqrt{0.5} \approx 0.707$$

$$\boxed{RMSE \approx 0.707}$$

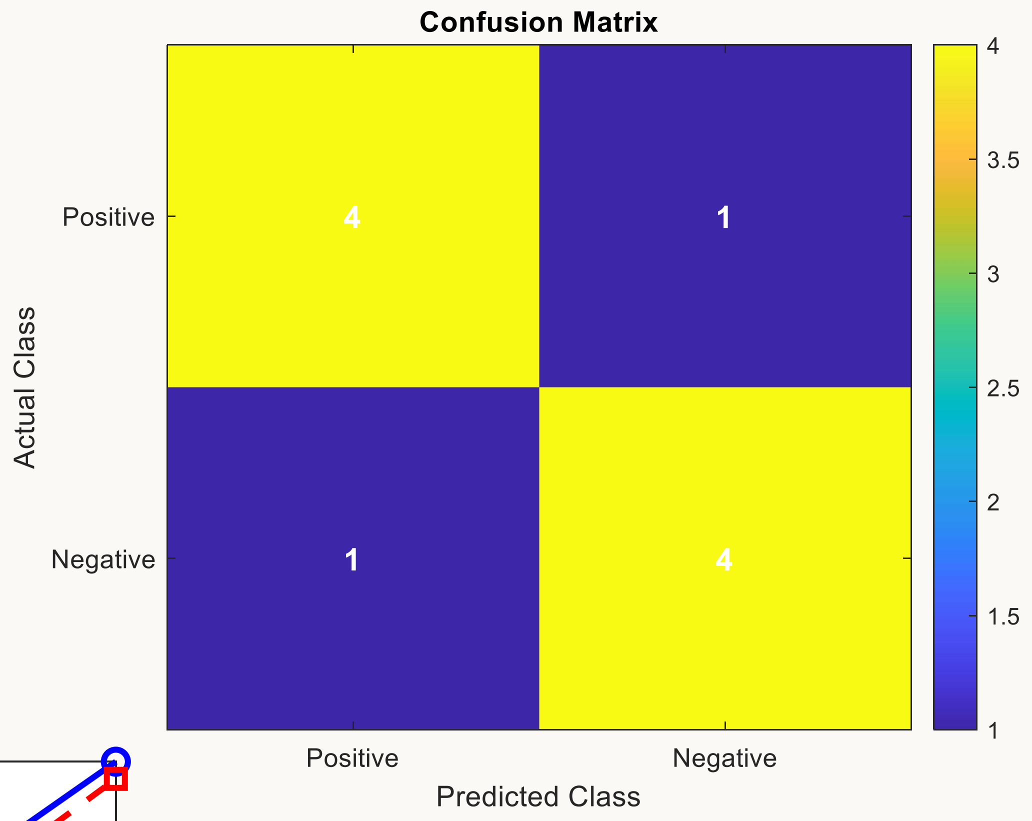


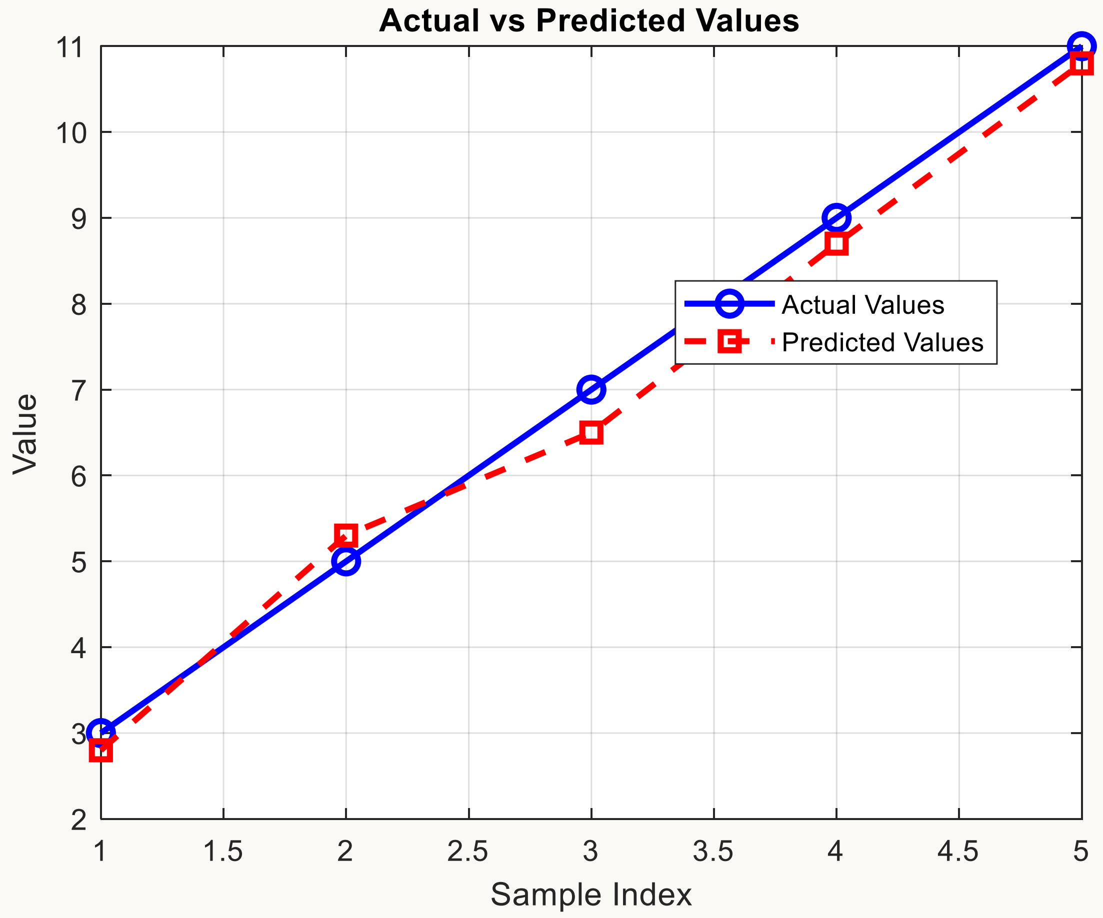


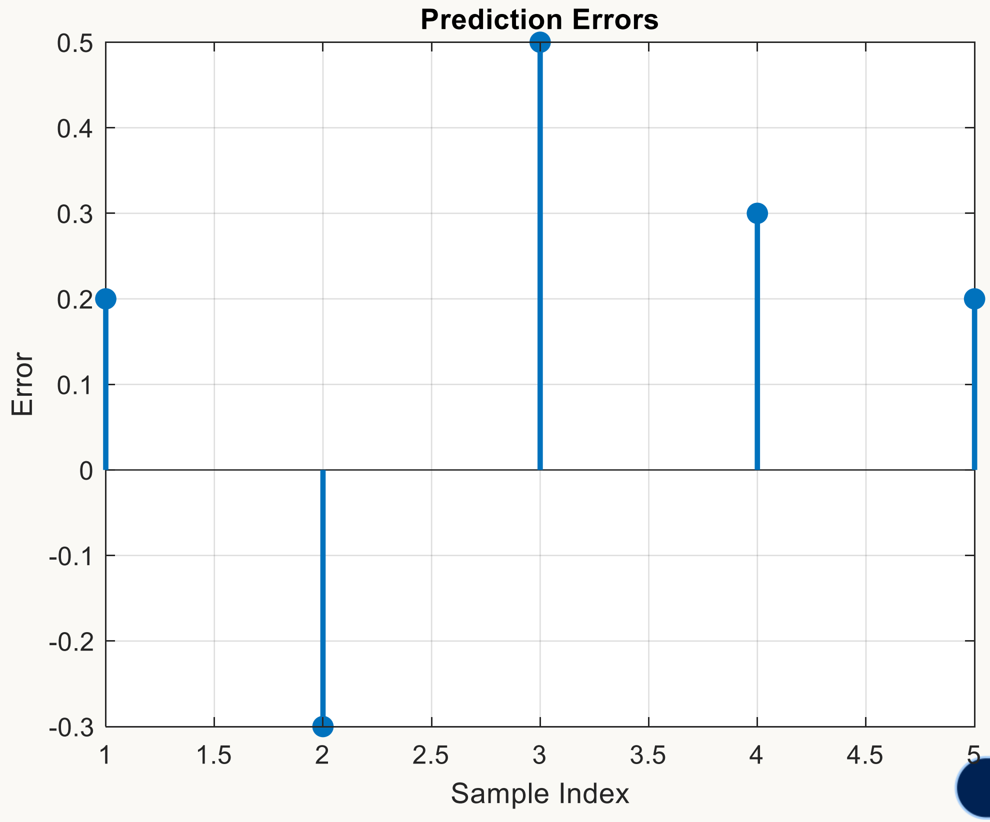
```
% Example regression data
y_actual = [3 5 7 9 11];
y_est    = [2.8 5.3 6.5 8.7 10.8];

% Convert to column vectors
y_actual = y_actual(:);
y_est    = y_est(:);
```

```
Regression Summary:
  MAE   = 0.3000
  MSE   = 0.1020
  RMSE  = 0.3194
  R^2   = 0.9872
```







Q3 (Regression Metrics)

A regression model predicts the target values $\hat{y}$for 4 samples.

Given:

$$y = [4,\ 6,\ 8,\ 10]$$
$$\hat{y} = [3,\ 7,\ 8.5,\ 9]$$

Required:

1. Compute the error vector $e = y - \hat{y}$.
2. Compute **MAE**, **MSE**, and **RMSE**.

Model Answer (Solution)

$$e = y - \hat{y} = [4-3,\ 6-7,\ 8-8.5,\ 10-9]$$
$$\boxed{e = [1,\ -1,\ -0.5,\ 1]}$$

$$MAE = \frac{|1| + |-1| + |-0.5| + |1|}{4}$$
$$MAE = \frac{1+1+0.5+1}{4} = \frac{3.5}{4} = 0.875$$
$$\boxed{MAE = 0.875}$$

$$MSE = \frac{(1)^2 + (-1)^2 + (-0.5)^2 + (1)^2}{4}$$
$$MSE = \frac{1+1+0.25+1}{4} = \frac{3.25}{4} = 0.8125$$
$$\boxed{MSE = 0.8125}$$

$$RMSE = \sqrt{MSE} = \sqrt{0.8125} \approx 0.901$$
$$\boxed{RMSE \approx 0.901}$$

Q4 (Classification Metrics)
A binary classifier produced the following confusion matrix results:

- $TP = 28$
- $TN = 52$
- $FP = 8$
- $FN = 12$

Required: Compute

1. **Accuracy**
2. **Precision**
3. **Recall**
4. **F1-score**

Solution
Total samples:

$$N = TP + TN + FP + FN = 28 + 52 + 8 + 12 = 100$$

$$Accuracy = \frac{TP + TN}{N} = \frac{28 + 52}{100} = \frac{80}{100} = 0.80$$
$$\boxed{Accuracy = 0.80 = 80\%}$$

$$Precision = \frac{TP}{TP + FP} = \frac{28}{28 + 8} = \frac{28}{36} = 0.7778$$
$$\boxed{Precision \approx 0.778}$$

$$Recall = \frac{TP}{TP + FN} = \frac{28}{28 + 12} = \frac{28}{40} = 0.70$$
$$\boxed{Recall = 0.70}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$
$$F1 = 2 \cdot \frac{0.7778 \cdot 0.70}{0.7778 + 0.70}$$
$$F1 = 2 \cdot \frac{0.5445}{1.4778} = 2 \cdot 0.3684 = 0.7368$$
$$\boxed{F1 \approx 0.737}$$

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